

SUMMARY TABLE: GENERAL SCIENCE GRADE 10

Sub-Strand 3.1: Turning Effect of Force — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 3.1: Turning Effect of Force
Driving Question	DRIVING QUESTION: Why does where you push, pull, or grip something determine how easy or hard it is to make it turn — and how do engineers and farmers in Kenya use this to design better tools?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 The Jembe Handle Puzzle — Launching the Phenomenon	Two learners — Amina and Juma — digging school-garden soil in Kericho using jembes with identical blade weights but different handle lengths. Amina (long handle ≈90 cm) finished quickly and easily; Juma (short handle ≈30 cm) struggled and tired rapidly. A ruler push-activity showed that pushing at the far end feels easier than pushing close to the pivot finger.	We have not yet learned the scientific explanation — we are still in the prediction stage. We noticed that handle length and grip position seem to matter. We generated predictions and questions. We created our first model sketch of force arrows and a pivot on a jembe. Our Driving Question Board is now open and holds all our current wonderings and questions.	Nothing has been formally explained yet — this is intentional. Our best current prediction is that gripping further from the metal blade makes digging easier, but we do not yet know WHY in scientific terms. We will investigate this in Lesson 2 using a metre rule and hanging masses to find the pattern.
Lesson 2 Moments of a Force at a Point — The Metre Rule Balancing Investigation	Learners balanced a metre rule on a pencil pivot using masses at measured distances on each side. They recorded weight (F in Newtons) and distance (d in cm/m) for at least five balanced configurations and computed $F \times d$ for both sides of the pivot in every trial.	For every balanced configuration, $F_1 \times d_1 = F_2 \times d_2$. This pattern is the Principle of Moments: a body is in rotational equilibrium when the sum of clockwise moments equals the sum of anticlockwise moments about the pivot. The moment of a force is $M = F \times d$ (units: N·m).	The Kericho jembe farmer's long handle works because the blade-soil contact point is the pivot. A longer handle increases the perpendicular distance d from the pivot to the line of effort. For the same effort force F, a larger d produces a larger moment $M = F \times d$ — meaning a greater turning effect on the blade — so the soil is broken with less physical effort from the farmer.
Lesson 3 Formalising the Moment Equation: $M = F \times d$ and the Principle of Moments	Learners watched an animation/simulation showing a seesaw with adjustable loads and pivot distances, and used digital devices or print media to research the definition, formula, and unit of the moment of a force.	The moment of a force $M = F \times d$ (unit: Newton-metre, Nm); the principle of moments (sum of clockwise moments = sum of anti-clockwise moments at equilibrium); the condition for rotational equilibrium; and that the Kericho farmer's	The jembe handle puzzle: the long-handled jembe is easier to use not because the force applied is larger, but because the perpendicular distance d from the grip to the blade pivot is larger, producing a much greater moment $M = F \times d$ at the blade —

		long-handled jembe produces a moment 5× greater than the short-handled jembe for the same applied force.	making it easier to rotate the blade through hard soil with the same muscular effort.
Lesson 4 Anti-Parallel Forces and Couples: When Two Pushes Make a Turn	Learners observed a suspended metre rule rotate when two equal, opposite (anti-parallel) forces were applied at different positions along it — even though the net translational force was zero. They observed that moving the two forces further apart caused greater rotation, and reducing the separation reduced the rotation.	Learners learned that a COUPLE is a pair of equal, parallel, antiparallel forces whose lines of action do not coincide. The moment of a couple = $F \times d$ (where d is the perpendicular distance between the two forces). A couple produces pure rotation with zero net translational force. Real-life Kenyan examples include a matatu steering wheel, a pipe wrench used by a Kisumu plumber, and the two-handed jembe grip.	The Kericho farmer's two-handed jembe grip forms a couple. By spreading her hands further apart (increasing d), she increases the moment of the couple ($\tau = F \times d$) without pushing harder (F unchanged). This greater rotational effect drives the blade into the soil more powerfully. This explains why widening the grip — or lengthening the handle — reduces the perceived effort needed to turn the jembe.
Lesson 5 Calculating Moments of Anti-Parallel Forces — Numerical Problems	Learners solved three graded numerical problems (jembe, matatu door, Rift Valley ox-cart) using $M = F \times d$, labelling each moment as clockwise or anticlockwise and calculating the net moment to determine the direction of rotation.	The turning effect (moment) of a force depends on BOTH the size of the force AND its perpendicular distance from the pivot. For anti-parallel forces, the moments act in opposite directions; the net moment determines which way the object rotates. This explains precisely — not just qualitatively — why a longer jembe handle makes digging easier.	When two anti-parallel forces act on an object, each creates a moment about the pivot: $M = F \times d$. If the anticlockwise moments exceed the clockwise moments, the object rotates anticlockwise — and vice versa. The farmer in Kericho produces a much larger anticlockwise moment with a long handle (large d) than the soil resistance produces as a clockwise moment close to the blade pivot, so the blade digs in with relatively little effort.
Lesson 6 Walk-Around and Real-Life Application Mapping: Turning Effect of Force in Our Environment	Learners walked around the school compound and identified real-life examples of turning effect of force — including the school gate (pivot at hinge), water tap handle (pivot at spindle), window stay scissor hinge, flagpole halyard cleat, and a groundskeeper's crowbar. They also observed video clips of a construction crane at Mombasa port, long-handled pliers at Kamukunji market in Nairobi, and a boda-boda bicycle brake lever in Kisumu. In every case, effort was applied at a distance from a fixed pivot, producing a turning effect.	The turning effect of force (moment) is not unique to the jembe — it is the same principle operating in gates, taps, pliers, cranes, and bicycle brakes across Kenya. Moment = Force $\times$ perpendicular distance from pivot. A longer effort arm produces a greater moment for the same applied force, making turning easier. Engineers and tool-makers deliberately design long handles and lever arms to multiply the turning effect and reduce the effort needed by the user.	The Principle of Moments explains why the farmer in Kericho grips the jembe far from the blade: the greater the perpendicular distance from the pivot (the blade-soil contact point), the greater the moment produced by the same muscular effort. This same principle is exploited by Kenyan engineers designing crane booms, jua-kali artisans choosing long-handled pliers, and boda-boda manufacturers sizing brake levers — all are maximising the moment arm to make turning forces more effective with less effort.
Lesson 7 DQB Update and Model Building: Unifying the Turning Effect of Force	Learners observed their own Lesson 1 jembe sketches placed alongside all evidence collected across six lessons — metre-rule data tables, couple investigation notes, and walk-around photographs — revealing the gap between their initial naive	Learners learned that a single equation, $M = F \times d$, unifies every turning-effect phenomenon encountered in the unit: the jembe, the school gate, the seesaw, the bicycle brake, and the crane all involve a pivot, a force, and a perpendicular distance,	Learners explained why the farmer in Kericho finds the long-handled jembe easier by constructing a fully labelled scientific diagram showing that the effort moment of 96 Nm far exceeds the load moment of 4 Nm, and by connecting this calculation to

	model and a scientifically complete explanation of the turning effect of force.	and increasing the distance always produces a larger moment for the same applied force.	the updated Driving Question Board answer: where you push or grip something determines the moment arm, and the moment arm determines the turning effect produced by that force.
Lesson 8 Consolidation, Assessment and Reflection: The Final Explanation of Turning Effect of Force	Learners observed: (i) their own Lesson 1 prediction cards alongside their current understanding, making conceptual change visible; (ii) the metre-rule balancing task, where a physical system achieves rotational equilibrium only when masses are placed at distances that satisfy the Principle of Moments; (iii) the anchoring phenomenon image (jembe handle puzzle) displayed for the final time, now viewed through a fully developed scientific lens.	Learners demonstrated that: the moment of a force equals the force multiplied by its perpendicular distance from the pivot ($M = F \times d$); a body is in rotational equilibrium when the sum of clockwise moments equals the sum of anti-clockwise moments; anti-parallel forces forming a couple produce a net turning effect equal to $F \times d$ with no resultant translational force; and the long-handled jembe is easier to use because the greater perpendicular distance from the pivot (blade-soil contact) produces a larger moment from the same applied force — fully explaining the anchoring phenomenon.	Learners constructed a three-sentence class Final Explanation: 'A moment is the turning effect of a force, calculated as the force multiplied by its perpendicular distance from the pivot. In the jembe, the blade-soil contact is the pivot — a longer handle increases the perpendicular distance, so the same effort from the farmer creates a much larger moment, making it easier to break hard soil. This is why engineers and farmers design tools with longer handles or wider grips: to maximise the turning effect without increasing the effort required.'

ATTRIBUTION AND LICENSING

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